



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-24
MONDAY TEST

CLASS : X
SUBJECT: MATHEMATICS

FULL MARKS: 40
DATE: 15.06.23

General Instructions:

- The paper consists of two printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

SECTION: A

[5 x 1 = 5]

1 i) The value(s) of k for which the quadratic equation $2x^2 - kx + k = 0$ has equal roots is (are)

- a) 4 b) 0 only c) 8 only d) 0, 8

ii) The quadratic equation $x^2 - 4x + 1 = 0$ has

- a) Equal roots b) Rational roots c) Irrational roots d) Imaginary roots

iii) If $x \in \mathbb{R}$, the solution set of $6 \leq -3(2x - 4) < 12$ is

- a) $\{x : x \in \mathbb{R}, 0 < x \leq 1\}$ b) $\{x : x \in \mathbb{R}, 0 \leq x < 1\}$
c) $\{0, 1\}$ d) $\{1\}$

iv) If Rakesh opened a recurring deposit account in a bank and deposited ₹800 per month for 1 year 6 months, then the total money deposited in the account is

- a) ₹11400 b) ₹14400 c) ₹13680 d) ₹14100

v) If the rate of CGST is 9% , then the rate of GST will be

- a) 9% b) 4.5% c) 18% d) 27%

SECTION: B

2. The marked price of an article is ₹18000 and the rate of GST on it is 12%. A shopkeeper buys this article at a reduced price and sells it at its marked price. If the shopkeeper paid ₹198.24 as CGST to the government, find the amount (inclusive of tax) paid by the shopkeeper . [4]

3. Mr Rakesh gets ₹6455 at the end of one year at the rate of 14% per annum in a Recurring Deposit Account. Find the monthly instalment. [4]

4. Solve the following equation for x and give your answer correct to 2 decimal places;
 $x^2 + 6x - 10 = 0$ [4]

5. i) Simplify; $\frac{-1}{3} \leq \frac{x}{2} - \frac{4}{3} < \frac{1}{6} : x \in \mathbb{R}$. Graph the value of x on the real number line.
 ii) Find the largest value of x for which $2(x - 1) \leq 9 - x$ and $x \in \mathbb{W}$. [3 + 3]
6. Find the value of m ; if the roots of the following quadratic equations are equal :
 $(4 + m)x^2 + (m + 1)x + 1 = 0$ [4]
7. A train travels a distance of 300 km at a constant speed. If the speed of the train is increased by 5km an hour, the journey would have taken 2 hours less. Find the speed of the train. [4]
8. Raghav deposited ₹200 per month for 36 months in a bank's recurring deposit account. If the bank pays interest at the rate of 11% per annum, find the amount he gets at maturity. [4]
9. For the following transaction of goods.

Item	Quantity	Rate per item(₹)	Discount	GST
A	1	2500	30%	18%
B	2	2000	20%	12%
C	1	1500	NIL	NIL

Calculate the :

i)Total taxable amount.

ii)Total bill amount.

[5]